AI Engineer Assessment

Finconecta

OVERVIEW & PURPOSE

The purpose of this assessment is to evaluate your ability to rapidly prototype, test, and validate hypotheses using various tools, integrations, and APIs. You will be assessed on your ability to creatively leverage AI tools and web scraping techniques to solve real-world challenges.

INSTRUCTIONS

This assessment includes both practical exercises and theoretical questions. The goal is to evaluate your ability to combine automation, AI, and scraping techniques for research and prototyping. You have **1 week** to complete the assessment.

DELIVERABLES

- Link to GitHub/GitLab repo for scripts.
- Document with a summary of your approach, challenges, and solutions.
- Optional: Screenshots or videos of working prototypes.

Technical Skills

- **Coding Challenge**
  Build an AI-powered application for a semantic search system:
    - Use a Generative AI API (e.g., OpenAI GPT or Cohere) to implement semantic search functionality.
    - Leverage a vector database (e.g., ChromaDB, FAISS, or Pinecone) to store and query embeddings.
    - Include basic functionalities: uploading documents, generating embeddings, and querying for similar documents.
- **Evaluation Criteria**:
    - Code readability and structure.
    - Effective use of vector databases and embedding techniques.
    - Performance optimization.

- **Web Scraping Task**
  Develop a web scraper to collect product data from an e-commerce site (e.g., product name, price, and description):
    - Use Python libraries like Scrapy or BeautifulSoup.
    - Deploy and automate the scraper using a scheduler (e.g., Cron jobs or AWS Lambda).
- **Evaluation Criteria**:
    - Scraper accuracy and error handling.
    - Use of best practices for automation and scalability.

- **Generative AI Workflow Task**
  Design a RAG pipeline for a Q&A system:
    - Use a pre-trained language model (e.g., OpenAI GPT).
    - Demonstrate tokenization and embeddings generation.
    - Integrate retrieval from a vector database to augment the generative model's response.
- **Evaluation Criteria**:
    - Understanding of RAG pipelines.
    - Optimization of token usage and cost efficiency.

    - Functional integration of retrieval and generation stages.

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2. Problem-Solving

- **Case Study**:
  FinConecta wants to improve customer engagement using an AI chatbot powered by Generative AI.
    - Describe how you would design and implement the solution.
    - Include details on the choice of APIs, scalability, security, and cost optimization strategies.
- **Evaluation Criteria**:
    - Clarity and feasibility of the solution.
    - Alignment with business needs.
    - Consideration of technical and operational challenges.

- **Debugging Task**:
  You're tasked to improve the latency of an AI agent deployed for automation. Identify potential bottlenecks in the system and propose solutions.
- **Evaluation Criteria**:
    - Ability to identify performance bottlenecks.
    - Practicality of proposed solutions.